Overview

JPS4+ is a Burst 2D Pathfinding solution tailored for 2D grid-based movement and pathing. It includes A* *(4 and 8-connected grids)* and JPS4 *(4-connected grids)* with horizontal/vertical canoncial ordering.

What is JPS4?

Jump Point Search Pathfinding in 4-connected Grids *(2025 Johannes Baum)* → Find it here

> *JPS4 builds upon the Jump Point Search (JPS8) algorithm, originally designed for 8-connected environments. To achieve efficient pathfinding on 4-connected grids, JPS4 employs a canonical ordering and a successor function that enable online graph pruning. This **reduces the search space by minimizing unnecessary node expansions.***
>
> *The core concept of JPS4 as well as JPS8 lies in the utilization of **jump points**. Strategically **placed at obstacle corners**, jump points prevent the search from overlooking crucial sections of the state space. They essentially reinitialize the canonical ordering, allowing exploration beyond obstacles. This mechanism ensures JPS4 finds optimal paths even in complex environments.*

This stands out from other 2D pathing solutions because it implements Jump Point Search Pathfinding in 4-connected Grids (Johannes Baum 2025) For dense maps, JPS4 can give >50x improvement v.s. A* by reducing the search space to remove unnecessary node expansions.

Resources

> → Asset Store Link

Algorithms

1. JPS4 Algorithm
2. A* Algorithm

Samples

1. Minimal Sample (2D)
2. Maze Sample (2D)
3. City Sample (3D)

Features

- A* (4-connected)
- A* (8-connected)
- Horizontal First JPS4 (4-connected)
- Vertical First JPS4 (4-connected)
- Movement Utils
- Animated Grid Movement
- Debugging in Editor